

DS200.Q21 - Bài thực hành 2

MSSV: 23520519

Bài 1

- Chạy lần lượt các lệnh sau:

```
cd "/mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai1"
pig -x local Code/Bai1.pig
```

```
hoang@DESKTOP-4HM890P:/mnt/d/UIT/DS200_Materials$ cd "/mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai1"
hoang@DESKTOP-4HM890P:/mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai1$ pig -x local Code/Bai1.pig
2026-04-16 17:42:36,475 INFO [main] pig.ExecTypeProvider (ExecTypeProvider.java:selectExecType(41)) - Trying ExecType : LOCAL
2026-04-16 17:42:36,477 INFO [main] pig.ExecTypeProvider (ExecTypeProvider.java:selectExecType(43)) - Picked LOCAL as the ExecType
2026-04-16 17:42:36,525 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-04-16 17:42:36,525 [main] INFO org.apache.pig.Main - Logging error messages to: /mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai1/pig_1776336156518.log
2026-04-16 17:42:36,543 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - user.name is deprecated. Instead, use mapreduce.job.user.name
2026-04-16 17:42:36,740 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file /home/hoang/.pigbootstrap not found
2026-04-16 17:42:36,810 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-04-16 17:42:36,811 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: file:///
2026-04-16 17:42:36,828 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-Bai1.pig-671e895b-9cf6-4b17-ad95-4d4d3a632311
2026-04-16 17:42:36,828 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-04-16 17:42:36,878 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - io.bytes.per.checksum is deprecated. Instead, use dfs.bytes-per-checksum
2026-04-16 17:42:36,880 [main] INFO org.apache.hadoop.fs.TrashPolicyDefault - Namenode trash configuration: Deletion interval = 0 minutes, Empty interval = 0 minutes.
Deleted KetQua
2026-04-16 17:42:37,146 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (G1 Old Gen) of size 1048576000 to monitor. collectionUsageThreshold = 734003200, usageThreshold = 734003200
2026-04-16 17:42:37,175 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-04-16 17:42:37,187 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: REPLICATED_JOIN,FILTER
2026-04-16 17:42:37,225 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInserter]}
2026-04-16 17:42:37,283 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-04-16 17:42:37,291 [main] WARN org.apache.hadoop.util.NativeCodeLoader - Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-04-16 17:42:37,307 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 2
2026-04-16 17:42:37,307 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 2
2026-04-16 17:42:37,334 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - session.id is deprecated. Instead, use dfs.metrics.session-id
2026-04-16 17:42:37,336 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Initializing JVM Metrics with processName=JobTracker, sessionId=
2026-04-16 17:42:37,356 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-04-16 17:42:37,362 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
```

Terminal khi chạy lệnh chứa thông tin user - Ảnh 1

```

Input(s):
Successfully read 131 records from: "file:///mnt/d/UIT/DS200_Materials/Assignment/Lab02/dataset/stopwords.txt"
Successfully read 11770 records from: "file:///mnt/d/UIT/DS200_Materials/Assignment/Lab02/dataset/hotel-review.csv"

Output(s):
Successfully stored 181204 records in: "file:///mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai1/KetQua"

Counters:
Total records written : 181204
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local100956016_0001 ->      job_local1861182460_0002,
job_local1861182460_0002

2026-04-16 17:42:39,279 [main] INFO  org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 17:42:39,280 [main] INFO  org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 17:42:39,281 [main] INFO  org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 17:42:39,285 [main] INFO  org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 17:42:39,286 [main] INFO  org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 17:42:39,287 [main] INFO  org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 17:42:39,293 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-04-16 17:42:39,303 [main] INFO  org.apache.pig.Main - Pig script completed in 3 seconds and 126 milliseconds (3126 ms)
hoang@DESKTOP-4HW890P:/mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai1$

```

Terminal khi chạy lệnh chứa thông tin user - Ảnh 2

- Kết quả:

Bài 2

- Chạy lần lượt các lệnh sau:

```

cd "/mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai2"
pig -x local Code/Bai2.pig

```

```

hoang@DESKTOP-4HW890P:/mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai2$ cd "/mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai2"
hoang@DESKTOP-4HW890P:/mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai2$ pig -x local Code/Bai2.pig
2026-04-16 17:54:20,935 INFO [main] pig.ExecTypeProvider (ExecTypeProvider.java:selectExecType(41)) - Trying ExecType : LOCAL
2026-04-16 17:54:20,937 INFO [main] pig.ExecTypeProvider (ExecTypeProvider.java:selectExecType(43)) - Picked LOCAL as the ExecType
2026-04-16 17:54:20,983 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-04-16 17:54:20,983 [main] INFO org.apache.pig.Main - Logging error messages to: /mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai2/pig_1776336860976.log
2026-04-16 17:54:21,002 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - user.name is deprecated. Instead, use mapreduce.job.user.name
2026-04-16 17:54:21,214 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file /home/hoang/.pigbootstrap not found
2026-04-16 17:54:21,292 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-04-16 17:54:21,293 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: file:///
2026-04-16 17:54:21,307 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-Bai2.pig-fe816505-188f-4d97-a63f-db3da3ea3fc4
2026-04-16 17:54:21,307 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-04-16 17:54:21,358 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - io.bytes.per.checksum is deprecated. Instead, use dfs.bytes-per-checksum
2026-04-16 17:54:21,360 [main] INFO org.apache.hadoop.fs.TrashPolicyDefault - Namenode trash configuration: Deletion interval = 0 minutes, Emptier interval = 0 minutes.
Deleted KetQua
2026-04-16 17:54:21,726 [main] WARN org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning IMPLICIT_CAST_TO_LONG 1 time(s).
2026-04-16 17:54:21,727 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-04-16 17:54:21,742 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP,BY,DISTINCT,FILTER
2026-04-16 17:54:21,786 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInserter]}
2026-04-16 17:54:21,812 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (G1 Old Gen) of size 1048576000 to monitor. collectionUsageThreshold = 734003200, usageThreshold = 734003200
2026-04-16 17:54:21,841 [main] INFO org.apache.pig.newplan.logical.rules.ColumnPruneVisitor - Columns pruned for data: $1, $4
2026-04-16 17:54:21,868 [main] WARN org.apache.hadoop.util.NativeCodeLoader - Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-04-16 17:54:21,884 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCCompiler - File concatenation threshold: 100 optimistic? false

```

Terminal khi chạy lệnh chứa thông tin user - Ảnh 1

```

Counters:
Total records written : 89
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local1760740014_0001
job_local11280794594_0002    ->    job_local11227307450_0003,job_local104726136_0004,
job_local11227307450_0003
job_local104726136_0004

2026-04-16 17:54:22,432 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 17:54:22,433 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 17:54:22,433 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 17:54:22,436 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 17:54:22,438 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 17:54:22,438 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 17:54:22,441 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 17:54:22,441 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 17:54:22,441 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 17:54:22,444 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 17:54:22,444 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 17:54:22,445 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 17:54:22,447 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-04-16 17:54:22,456 [main] INFO org.apache.pig.Main - Pig script completed in 1 second and 861 milliseconds (1861 ms)
hoang@DESKTOP-4HW890P:/mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai2$

```

Terminal khi chạy lệnh chứa thông tin user - Ảnh 2

- Kết quả:
 - Thống kê tần số xuất hiện của các từ. Chỉ ra các từ xuất hiện trên 500 lần:

```

1 1,672
2 qua,575
3 vui,746
4 ăn,1142
5 cùng,511
6 cả,1181
7 gian,627
8 già,2318
9 hải,1366
10 mái,507
11 nghỉ,1949
12 như,1302
13 sinh,1536
14 sự,598
15 tri,1745
16 về,675
17 vệ,1535
18 vì,1607
19 vu,3741
20 đã,1292

```

```

initialized
2026-04-16 17:54:22,444 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 17:54:22,444 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 17:54:22,445 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 17:54:22,447 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-04-16 17:54:22,456 [main] INFO org.apache.pig.Main - Pig script completed in 1 second and 861 milliseconds (1861 ms)
hoang@DESKTOP-4H4890P: /mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai2$ ~

```

- Thống kê số bình luận theo từng phân loại (category):

```

1 HOTEL,3086
2 ROOMS,1791
3 SERVICE,2298
4 LOCATION,956
5 FACILITIES,416
6 FOOD&DRINKS,543
7 ROOM_AMENITIES,1156
8

```

```

initialized
2026-04-16 17:54:22,444 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 17:54:22,444 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 17:54:22,445 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 17:54:22,447 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-04-16 17:54:22,456 [main] INFO org.apache.pig.Main - Pig script completed in 1 second and 861 milliseconds (1861 ms)
hoang@DESKTOP-4H4890P: /mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai2$ ~

```

- Thống kê số bình luận theo từng khía cạnh đánh giá (aspect):

```

1 PRICES,740
2 COMFORT,1342
3 GENERAL,4667
4 QUALITY,1091
5 CLEANLINESS,1304
6 MISCELLANEOUS,328
7 STYLE&OPTIONS,265
8 DESIGN&FEATURES,882

```

```

initialized
2026-04-16 17:54:22,444 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 17:54:22,444 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 17:54:22,445 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 17:54:22,447 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-04-16 17:54:22,456 [main] INFO org.apache.pig.Main - Pig script completed in 1 second and 861 milliseconds (1861 ms)
hoang@DESKTOP-4H4890P: /mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai2$ ~

```

Bài 3

- Chạy lần lượt các lệnh sau:

```
cd "/mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai3"
pig -x local Code/Bai3.pig
```

```

Success!

Job Stats (time in seconds):
JobId  Maps  Reduces  MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime  AvgReduceTime  MedianReducetime  A
lias  Feature  Outputs
job_local1141967165_0004  1  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  n/a  f
job_local1209305330_0007  1  0  n/a  n/a  n/a  n/a  0  0  0  0  f
job_local1220067722_0003  1  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  n/a  f
job_local1309324973_0009  1  0  n/a  n/a  n/a  n/a  0  0  0  0  f
file:///mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai3/KetQua/Aspect_Pos_NhieuNhat,  0  0  0  0  0  0  0  0  0  0  f
job_local1614067156_0008  1  0  n/a  n/a  n/a  n/a  0  0  0  0  f
file:///mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai3/KetQua/Aspect_NegSentiment_NhieuNhat,  0  0  0  0  0  0  0  0  0  0  f
job_local1976352164_0002  1  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  n/a  f
job_local1278075796_0001  1  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  n/a  f
a,raw_data  MULTI_QUERY,COMBINER
job_local1567482618_0005  1  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  n/a  f
job_local1851283526_0006  1  0  n/a  n/a  n/a  n/a  0  0  0  0  f

Input(s):
Successfully read 11770 records from: "file:///mnt/d/UIT/DS200_Materials/Assignment/Lab02/dataset/hotel-review.csv"

Output(s):
Successfully stored 1 records in: "file:///mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai3/KetQua/Aspect_NegSentiment_NhieuNhat"
Successfully stored 1 records in: "file:///mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai3/KetQua/Aspect_Pos_NhieuNhat"

Counters:
Total records written : 2
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local1278075796_0001 -> job_local1220067722_0003,job_local1976352164_0002,
job_local1220067722_0003 -> job_local1567482618_0005,
job_local1567482618_0005 -> job_local1851283526_0006,
job_local1851283526_0006 -> job_local1614067156_0008,
job_local1614067156_0008 -> job_local1141967165_0004,
job_local1976352164_0002 -> job_local1209305330_0007,
job_local1141967165_0004 -> job_local1309324973_0009,
job_local1209305330_0007 -> job_local1309324973_0009,
job_local1309324973_0009

```

Terminal khi chạy lệnh chứa thông tin user - Ảnh 1

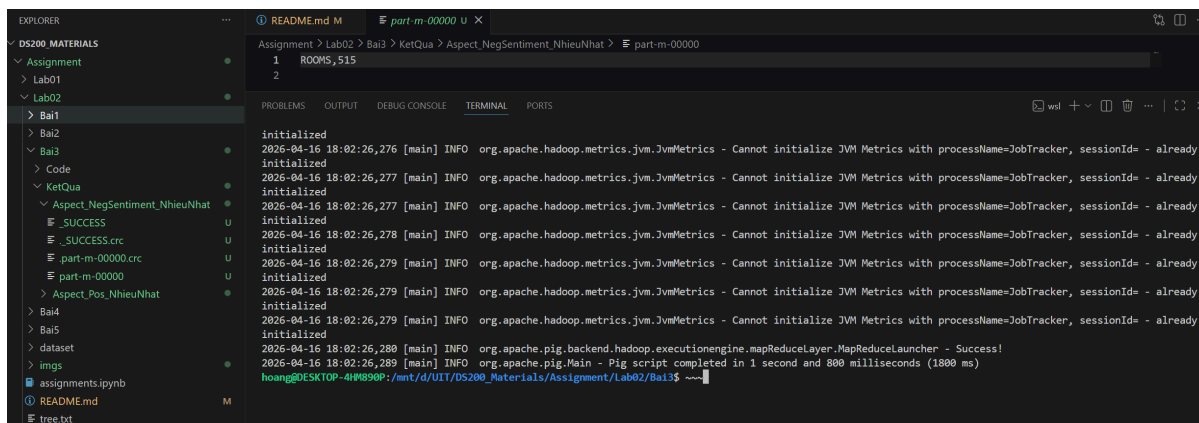
```

2026-04-16 18:02:26,266 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 18:02:26,266 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 18:02:26,268 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 18:02:26,269 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 18:02:26,269 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 18:02:26,270 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 18:02:26,271 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 18:02:26,271 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 18:02:26,273 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 18:02:26,273 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 18:02:26,274 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 18:02:26,276 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 18:02:26,276 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 18:02:26,276 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 18:02:26,277 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 18:02:26,277 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 18:02:26,278 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 18:02:26,279 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 18:02:26,279 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 18:02:26,279 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already
initialized
2026-04-16 18:02:26,280 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-04-16 18:02:26,289 [main] INFO org.apache.pig.Main - Pig script completed in 1 second and 800 milliseconds (1800 ms)
hoang@DESKTOP-4HM896P:/mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai3$

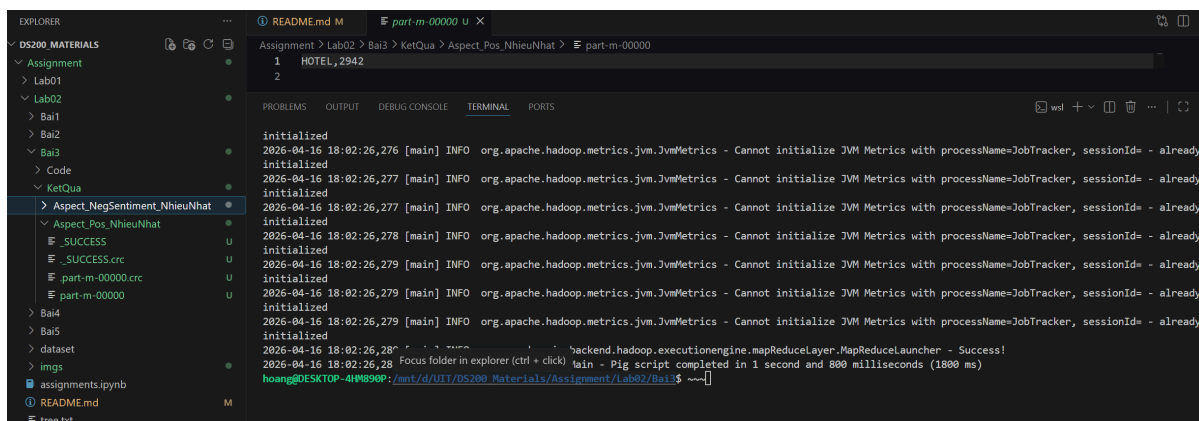
```

Terminal khi chạy lệnh chứa thông tin user - Ảnh 2

- Kết quả:
 - Xác định khía cạnh nào nhận nhiều đánh giá tiêu cực (negative) nhất:



- Xác định khía cạnh nào nhận nhiều đánh giá tích cực nhất (positive) nhất:



Bài 4

- Chạy lần lượt các lệnh sau:

```
cd "/mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai4"
pig -x local Code/Bai4.pig
```

```
HadoopVersion  PigVersion  UserId  StartedAt  FinishedAt  Features
2.7.3  0.17.0  hoang  2026-04-16 18:07:05  2026-04-16 18:07:07  REPLICATED_JOIN, GROUP_BY, FILTER

Success!

Job Stats (time in seconds):
JobId  Maps  Reduces  MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime  AvgReduceTime  MedianReduceTime  A
lias  Feature Outputs
job_local1423657051_0008  1  1  n/a  n/a  n/a  n/a  n/a  n/a  cleaned_words, data_final, joined_data, lower_data,
neg_counts, neg_data, neg_group, pos_counts, pos_data, pos_group, raw_data, words_flat  REPLICATED_JOIN, MULTI_QUERY, COMBINER
job_local1785288199_0007  1  0  n/a  n/a  n/a  n/a  0  0  0  stopwords  MAP_ONLY
job_local1405818273_0009  1  1  n/a  n/a  n/a  n/a  n/a  n/a  pos_ordered, sorted, top5_pos  GROUP_BY  file:///
mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai4/KetQua/Word_Pos_Top5,
job_local1915300588_0010  1  1  n/a  n/a  n/a  n/a  n/a  n/a  neg_ordered, sorted, top5_neg  GROUP_BY  file:///
mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai4/KetQua/Word_Neg_Top5,

Input(s):
Successfully read 131 records from: "file:///mnt/d/UIT/DS200_Materials/Assignment/Lab02/dataset/stopwords.txt"
Successfully read 11770 records from: "file:///mnt/d/UIT/DS200_Materials/Assignment/Lab02/dataset/hotel-review.csv"

Output(s):
Successfully stored 8 records in: "file:///mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai4/KetQua/Word_Pos_Top5"
Successfully stored 8 records in: "file:///mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai4/KetQua/Word_Neg_Top5"

Counters:
Total records written : 16
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local1785288199_0007  ->  job_local1423657051_0008,
job_local1423657051_0008  ->  job_local1405818273_0009, job_local1915300588_0010,
job_local1405818273_0009
job_local1915300588_0010
```

Terminal khi chạy lệnh chứa thông tin user - Ảnh 1

```

2026-04-16 18:07:07,749 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 18:07:07,749 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 18:07:07,750 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 18:07:07,751 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 18:07:07,752 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 18:07:07,752 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 18:07:07,753 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 18:07:07,754 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 18:07:07,754 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 18:07:07,758 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 18:07:07,758 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 18:07:07,758 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 18:07:07,758 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 18:07:07,759 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-04-16 18:07:07,767 [main] INFO org.apache.pig.Main - Pig script completed in 7 seconds and 960 milliseconds (7960 ms)
hoang@DESKTOP-4HMB90P:/mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai4$

```

Terminal khi chạy lệnh chứa thông tin user - Ảnh 2

- Kết quả:

- Dựa vào từng phân loại bình luận, hãy xác định **5 từ** mang ý nghĩa tích cực nhất:

The screenshot displays a VS Code workspace. The Explorer sidebar on the left shows a directory tree for 'DS200_MATERIALS'. The main editor area shows a file named 'part-r-00000' with the following content:

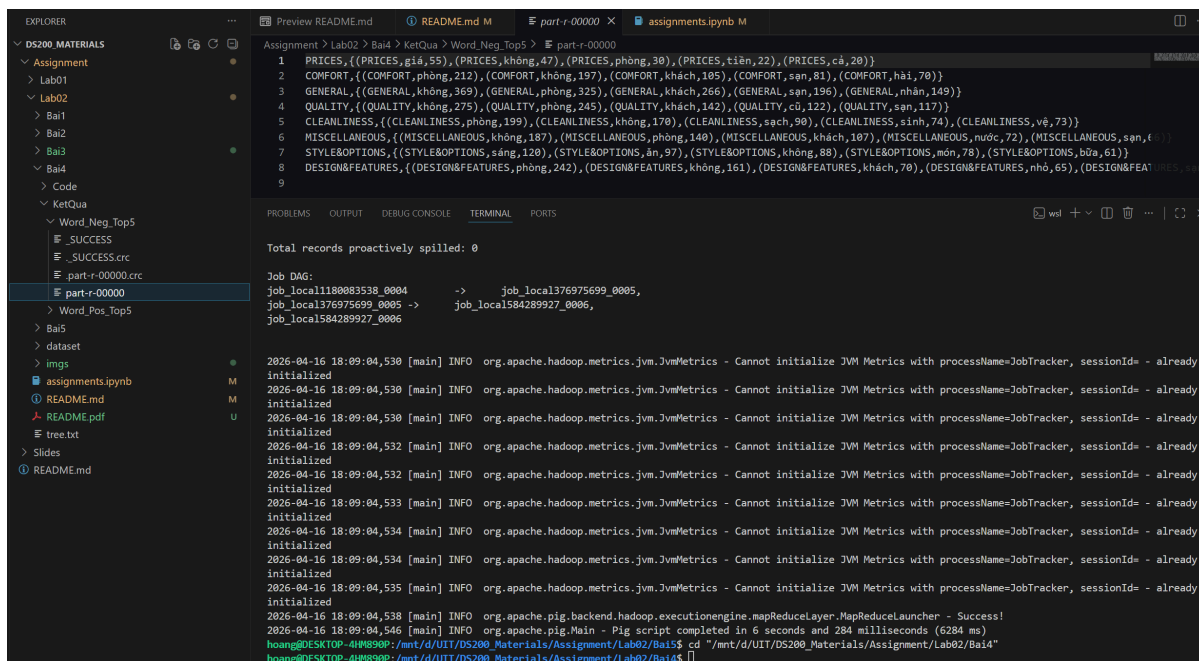
```

1 PRICES,{{(PRICES,gia,600),(PRICES,cá,344),(PRICES,vụ,298),(PRICES,hop,297),(PRICES,phong,269)}}
2 COMFORT,{{(COMFORT,hai,530),(COMFORT,làng,463),(COMFORT,phong,434),(COMFORT,khách,433),(COMFORT,san,365)}}
3 GENERAL,{{(GENERAL,khách,2223),(GENERAL,phong,2232),(GENERAL,vụ,1989),(GENERAL,nhân,1848),(GENERAL,viên,1884)}}
4 QUALITY,{{(QUALITY,phong,330),(QUALITY,chất,326),(QUALITY,lượng,297),(QUALITY,khách,267),(QUALITY,san,227)}}
5 CLEANLINESS,{{(CLEANLINESS,phong,1181),(CLEANLINESS,sạch,990),(CLEANLINESS,tiền,543),(CLEANLINESS,đc,495),(CLEANLINESS,vệ,445)}}
6 MISCELLANEOUS,{{(MISCELLANEOUS,khách,26),(MISCELLANEOUS,san,20),(MISCELLANEOUS,phong,17),(MISCELLANEOUS,không,10),(MISCELLANEOUS,án,10)}}
7 STYLEOPTIONS,{{(STYLEOPTIONS,sang,93),(STYLEOPTIONS,ăn,74),(STYLEOPTIONS,bữa,52),(STYLEOPTIONS,món,46),(STYLEOPTIONS,phong,36)}}
8 DESIGN&FEATURES,{{(DESIGN&FEATURES,phong,529),(DESIGN&FEATURES,dep,233),(DESIGN&FEATURES,sạch,230),(DESIGN&FEATURES,không,227),(DESIGN&FEATURES,phong,227)}}
9

```

The terminal at the bottom shows the output of a Pig script, including the success message and the completion time.

- Dựa vào từng phân loại bình luận, hãy xác định **5 từ** mang ý nghĩa tiêu cực nhất:



Bài 5

- Chạy lần lượt các lệnh sau:

```
cd "/mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai5"
pig -x local Code/Bai5.pig
```

```
hoang@DESKTOP-4HW890P:/mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai4$ cd "/mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai5"
hoang@DESKTOP-4HW890P:/mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai5$ pig -x local Code/Bai5.pig
2026-04-16 18:08:58,583 INFO [main] org.apache.pig.ExecTypeProvider (java:selectExecType(41)) - Trying ExecType : LOCAL
2026-04-16 18:08:58,584 INFO [main] org.apache.pig.ExecTypeProvider (java:selectExecType(43)) - Picked LOCAL as the ExecType
2026-04-16 18:08:58,634 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-04-16 18:08:58,634 [main] INFO org.apache.pig.Main - Logging error messages to: /mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai5/pig_1776337738628.log
2026-04-16 18:08:58,656 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - user.name is deprecated. Instead, use mapreduce.job.user.name
2026-04-16 18:08:58,877 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file /home/hoang/.pigbootstrap not found
2026-04-16 18:08:58,947 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-04-16 18:08:58,947 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: file:///
2026-04-16 18:08:58,962 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-Bai5.pig-8e67f22a-99f2-47c2-a35d-a035edc5abf4
2026-04-16 18:08:58,962 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-04-16 18:08:59,008 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - io.bytes.per.checksum is deprecated. Instead, use dfs.bytes-per-checksum
2026-04-16 18:08:59,009 [main] INFO org.apache.hadoop.fs.TrashPolicyDefault - Namenode trash configuration: Deletion interval = 0 minutes, Emptier interval = 0 minutes.
Deleted KetQua
2026-04-16 18:08:59,288 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (G1 Old Gen) of size 1048576000 to monitor. collectionUsageThreshold = 734003200, usageThreshold = 734003200
2026-04-16 18:08:59,304 [main] WARN org.apache.hadoop.util.NativeCodeLoader - Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-04-16 18:08:59,408 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: REPLICATED_JOIN, GROUP_BY, FILTER
2026-04-16 18:08:59,441 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInserter]}
2026-04-16 18:08:59,458 [main] INFO org.apache.pig.newplan.logical.rules.ColumnPruneVisitor - Columns pruned for raw_data: $0, $3, $4
2026-04-16 18:08:59,506 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-04-16 18:08:59,514 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-04-16 18:08:59,528 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 3
2026-04-16 18:08:59,528 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 3
2026-04-16 18:08:59,558 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - session.id is deprecated. Instead, use dfs.metrics.session-id
2026-04-16 18:08:59,560 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Initializing JVM Metrics with processName=JobTracker, sessionId=
2026-04-16 18:08:59,578 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-04-16 18:08:59,583 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-04-16 18:08:59,584 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent
```

Terminal khi chạy lệnh chứa thông tin user - Ảnh 1


```

2026-04-16 18:09:04,530 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 18:09:04,530 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 18:09:04,530 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 18:09:04,532 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 18:09:04,532 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 18:09:04,533 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 18:09:04,534 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 18:09:04,534 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 18:09:04,535 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-16 18:09:04,538 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-04-16 18:09:04,546 [main] INFO org.apache.pig.Main - Pig script completed in 6 seconds and 284 milliseconds (6284 ms)
hoang@DESKTOP-4HM890P:/mnt/d/UIT/DS200_Materials/Assignment/Lab02/Bai5$

```

Terminal khi chạy lệnh chứa thông tin user - Ảnh 2

- Kết quả:

